Ahmedabad University

CSE623 Machine Learning: Theory and Practice

Topic:
Machine Learning Based Risk Parity Portfolio Construction for Financial Markets

Weekly Report 7

Submitted to: Prof. Mehul Raval
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Team Name: RiskNeutral
Team Members:

Bhavya Sharma	*AU24110006*	*Master of Science in Quantitative Finance*
Ritik Rakesh	*AU24110010*	*Master of Science in Quantitative Finance*
Smit Kalathiya	*AU24110013*	*Master of Science in Quantitative Finance*
Shree S	*AU24110008*	*Master of Science in Quantitative Finance*
Vivek Maskar	*AU24110011*	*Master of Science in Quantitative Finance*

1. **Overview of Project:**

Risk parity has emerged as a widely adopted portfolio construction framework that allocates capital by equalizing risk contributions across assets rather than maximizing expected returns. Popularized in institutional asset management as an "all weather" strategy, risk parity portfolios are designed to deliver stable performance across different market regimes. However, the effectiveness of risk parity critically depends on accurate estimation of asset volatilities and correlations.

Recent advances in machine learning and statistical learning theory provide tools for denoising and dynamically estimating covariance structures. Techniques such as shrinkage estimators, random matrix theory based eigenvalue filtering, and hierarchical clustering provide more robust representations of dependency structures (tail risks) for a multi-asset portfolio.

This project builds a risk parity portfolio using machine learning techniques. Using daily US financial assets data from 2015–2025, we construct long-short and long-only portfolios using PCA-based risk parity and Hierarchical Risk Parity (HRP). We evaluate their out-of-sample performance from 2020 to 2025 and compare them against the S&P 500 benchmark.

2. **Overview of Week 7:**

Tasks accomplished this week:

- Conducted a comprehensive side-by-side comparison of PCA-based risk parity (from Week 4) and Hierarchical Risk Parity across all performance metrics, using the same dataset, rolling window parameters, and evaluation period for a fair comparison.
- Documented the complete mathematical framework for HRP, including the correlation-to-distance transformation, Ward linkage, quasi-diagonalization, and recursive bisection with full derivations and a three-asset worked example.
- Identified the prerequisite mathematical topics required to understand the HRP pipeline and organized them into a structured study guide covering linear algebra, probability, clustering, portfolio theory, and recursive algorithms.
- Finalized the codebase, added documentation, and uploaded to GitHub. Completed the final IEEE conference-style report.

3. **Work Completed:**

PCA+GMV vs. HRP: Side-by-Side Comparison

- Both pipelines were evaluated on the same out-of-sample period (2020–2025) using Marčenko–Pastur denoised covariance, 4-year training window, and 21-day holding period. The PCA pipeline uses eigen decomposition, ridge regularization ($\lambda=10^{-3}$), and the closed-form GMV formula $w = \Sigma^{-1}1/(1^T\Sigma^{-1}1)$, producing long-short net-long weights. The HRP pipeline uses hierarchical clustering (Ward), quasi-diagonalization, and recursive bisection, producing long-only weights with no matrix inversion.
- Key comparison findings: (i) HRP produces naturally bounded weights without requiring ridge regularization or an explosion guard; (ii) HRP's long-only structure avoids the negative-return issue encountered in Week 3 when long-only constraints were imposed on the PCA pipeline's inherently

signed factor exposures; (iii) HRP demonstrates more stable weight evolution across rolling windows, with lower turnover.

Mathematical Framework Documentation

- A comprehensive mathematical framework document was prepared covering the three phases of HRP. Phase 1 (Tree Clustering) derives the distance metric $d_{ij} = \sqrt{(1-\rho_{ij})/2}$, explains Ward's linkage criterion, and describes the dendrogram construction. Phase 2 (Quasi-Diagonalization) formalizes the permutation $\tilde{\Sigma} = P\Sigma P^T$ and its effect on the covariance structure. Phase 3 (Recursive Bisection) derives the IVP weights $w_i = (1/\Sigma_{ii})/\Sigma(1/\Sigma_{jj})$, the cluster variance $V = w^T\Sigma w$, and the split factor $\alpha = V\_R/(V\_L+V\_R)$.
- A complete numeric example with three assets (Tech Stock 1, Tech Stock 2, Government Bond) was worked through, demonstrating both iterations of the bisection and verifying that final weights sum to 100%.

Prerequisite Study Guide

- Ten prerequisite mathematical topics were identified and organized in a progressive study sequence: (1) Linear Algebra (quadratic forms, PSD, eigenvalues), (2) Probability and Statistics (covariance, sample estimation), (3) Covariance Denoising (shrinkage, RMT), (4) Metric Spaces (triangle inequality, correlation distance), (5) Hierarchical Clustering (Ward linkage, dendrograms, chaining problem), (6) Graph Theory (binary trees, traversal), (7) Portfolio Theory (Markowitz, GMV, IVP, risk budgeting), (8) Estimation Error (condition number, ridge), (9) Recursive Algorithms (divide-and-conquer, $O(N^2 \log N)$ complexity), (10) Time Series Backtesting (stationarity, rolling windows).

Code Finalization and GitHub Upload

- The codebase was organized into two main notebooks: the PCA Risk Parity notebook (Weeks 2–4) and the HRP Risk Parity notebook (Weeks 5–7). Both notebooks share the same data preparation, covariance estimation, and performance evaluation infrastructure. All functions are documented with docstrings explaining the mathematical formulation, input/output types, and references.
- The code was uploaded to GitHub with a README documenting the project structure, dependencies, data requirements, and instructions for reproducing the results.

4. Challenges Encountered:

- Comparing PCA+GMV (long-short) with HRP (long-only) fairly is inherently difficult because the two pipelines operate under different constraints. The PCA pipeline can take short positions to hedge, while HRP cannot. A direct Sharpe ratio comparison may favor whichever approach happens to suit the specific market regime of the evaluation period.
- Ensuring that the mathematical framework document correctly represents both López de Prado's original formulation and our Ward linkage adaptation required careful cross-referencing of the 2016 paper, the 2018 book (Advances in Financial Machine Learning), and the Gemini-provided derivation. One discrepancy was identified: López de Prado uses single linkage while we use Ward, which we justified on empirical grounds.

- Condensing seven weeks of iterative development into a coherent 2-page IEEE report required significant editorial judgment about which details to include and which to leave for the supplementary materials.

5. **Summary and Conclusion:**

- Over the seven-week project, we built a complete machine learning-based risk parity pipeline from data collection through out-of-sample evaluation. The project explored two complementary approaches: PCA-based factor risk parity (requiring regularized matrix inversion) and HRP (requiring no inversion). Both approaches were evaluated across three covariance estimators and a five-year out-of-sample period.
- HRP emerged as the more practically robust approach, producing naturally bounded, long-only weights without the regularization and guard mechanisms required by the PCA pipeline. The Marčenko–Pastur denoised covariance provided the best input for both pipelines, confirming that covariance estimation quality is the critical determinant of portfolio construction success.
- Future extensions include: implementing dynamic regime detection to adapt the training window, exploring non-hierarchical clustering methods (e.g., HDBSCAN), and testing on international multi-asset universes beyond U.S. markets.

References:

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